

①

$$V_c = V_1(1 - e^{-t/RC})$$

$$1/3 = e^{-t/RC}$$

$$2/3 V_1 = V_1(1 - e^{-t/RC}) \quad t/RC = \ln(3)$$

$$2/3 = 1 - e^{-t/RC}$$

$$t = \ln(3)RC$$

$$2/3 - 1 = -e^{-t/RC}$$

$$\underline{t = RC \ln(3)}$$

②

$$1/3 V_1 = V_1(1 - e^{-t/RC}) \quad t = RC \ln(3) -$$

$$1/3 - 1 = -e^{-t/RC}$$

$$(-\ln(2/3)RC)$$

$$\ln(2/3) = -\frac{t}{RC}$$

$$t = RC \ln(3) + \ln(2/3)$$

$$-t = \ln(2/3)RC \quad RC$$

$$t = -\ln(2/3)RC$$

$$\underline{t = RC \ln(2)}$$

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$$0.001 V_i = \frac{2}{3} V_i e^{-t/RC}$$

$$0.003/2 = e^{-t/RC}$$

$$0.0015 = e^{-t/RC}$$

$$\ln(0.0015) = -\frac{t}{RC} \quad \text{Tiende a 0}$$

$$\underline{t = -RC(\ln(0.0015))}$$